

Sudo apt Install git & Sudo apt update

```
dyp@dyp-OptiPlex-3000:~$ git
Command 'git' not found, but can be installed with:

sudo apt install git

dyp@dyp-OptiPlex-3000:~$ sudo apt install git
[sudo] password for dyp:
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 3513 (unattended-upgr)
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  chronun-codecs-ffmpeg-extra gstreamer1.0-vaapi libgstreamer-plugins-bad1.0-0 libva-wayland2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-el git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 67 not upgraded.
Need to get 5,525 kB of archives.
After this operation, 38.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu focal/main amd64 liberror-perl all 0.17029-1 [26.5 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 git-man all 1:2.25.1-1ubuntu3.13 [687 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 git amd64 1:2.25.1-1ubuntu3.13 [4,612 kB]
Fetched 5,525 kB in 7s (737 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 180140 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17029-1_all.deb ...
Unpacking liberror-perl (0.17029-1) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1:2.25.1-1ubuntu3.13_all.deb ...
Unpacking git-man (1:2.25.1-1ubuntu3.13) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1:2.25.1-1ubuntu3.13_amd64.deb ...
Unpacking git (1:2.25.1-1ubuntu3.13) ...
Setting up liberror-perl (0.17029-1) ...
Setting up git-man (1:2.25.1-1ubuntu3.13) ...
Setting up git (1:2.25.1-1ubuntu3.13) ...
Processing triggers for man-db (2.9.1-1) ...
dyp@dyp-OptiPlex-3000:~$ sudo apt update
Hit:1 http://dell.archive.canonical.com focal InRelease
Hit:2 http://security.ubuntu.com/ubuntu focal-security InRelease
```

Git congfig -- global User.name (your name)

Git congfig -- global User.email (your email)

Git congfig --list

```
git help -a' and 'git help -g' list available subcommands and some
concept guides. See 'git help <command>' or 'git help <concept>'
to read about a specific subcommand or concept.
See 'git help git' for an overview of the system.
dyp@dyp-OptiPlex-3000:~$ git --version
git version 2.25.1
dyp@dyp-OptiPlex-3000:~$ git config --global user.name "Pavan"
dyp@dyp-OptiPlex-3000:~$ git config --global user.email "pp8980929"
dyp@dyp-OptiPlex-3000:~$ git config --list
user.name=Pavan
user.email=pp8980929
dyp@dyp-OptiPlex-3000:~$ git config --global user.email "pp8980929@gmail.com"
dyp@dyp-OptiPlex-3000:~$ git config --list \
- git config --list \
.
error: wrong number of arguments, should be 0
usage: git config [<options>]

config file location

--global          use global config file
--system          use system config file
--local           use repository config file
--worktree        use per-worktree config file
-f, --file <file> use given config file
--blob <blob-id>  read config from given blob object

action

--get             get value: name [value-regex]
--get-all        get all values: key [value-regex]
--get-regexp      get values for regexp: name-regex [value-regex]
--get-urlmatch    get value specific for the URL: section[.var] URL
--replace-all     replace all matching variables: name value [value-regex]
--add            add a new variable: name value
--unset          remove a variable: name [value-regex]
--unset-all      remove all matches: name [value-regex]
--rename-section  rename section: old-name new-name
--remove-section  remove a section: name

-l, --list        list all
-e, --edit        open an editor
--get-color       find the color configured: slot [default]
--get-colorbool   find the color setting: slot [stdout-is-ty]

type

-t, --type <v>  value is given this type
--bool          value is "true" or "false"
--int           value is decimal number
```

Create new repository

(git clone <https://github.com/use>name/repository name/file name {path})

```
core.logallIUpdates=true
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject$ git clone https://github.com/pavyaa87/experiment-no-1-
Cloning into 'experiment-no-1'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

Cd repository name

ls

Cat README.md

Touch file_name

ls

Gedit python

```
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject$ git clone https://github.com/pavyaa87/experiment-no-1-
Cloning into 'experiment-no-1'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject$ cd experiment-no-1
bash: cd: experiment-no-1: No such file or directory
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject$ cd experiment-no-1
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject/experiment-no-1$ ls
README.md
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject/experiment-no-1$ cat README.md
# experiment-no-1-
this is new dir
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject/experiment-no-1$ touch python.py
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject/experiment-no-1$ ls
python.py README.md
mr-pravin@mr-pravin-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/myproject/experiment-no-1$ gedit python.py
Command 'gedit' not found, but can be installed with:
sudo snap install gedit # version 46.1, or
sudo apt install gedit # version 46.1-3
See 'snap info gedit' for additional versions.
```

Create a new branch

Git Branch

Git branch (branch name)

```
sudo apt install genomertools
```

```
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git branch
* main
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git branch snow
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git branch
* main
snow
```

Git Status

```
dyp@dyp-OptiPlex-3000: ~/AsB_43/devops
no changes added to commit (use "git add" and/or "git commit -a")
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
(Use "git pull" to update your local branch)
```

Git Add

Git Commit -m

```
dyp@dyp-OptiPlex-3000: ~/AsB_43/devops
(use "git restore <file>..." to discard changes in working directory)
modified:   hello1.py

no changes added to commit (use "git add" and/or "git commit -a")
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git add .
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git commit -m "cc"
> ^C
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git commit -m "cc"
[main b0384ed] cc
1 file changed, 3 insertions(+), 4 deletions(-)
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ ls
hello1.py hello1.ty README.md
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git status
On branch main
Your branch and 'origin/main' have diverged,
and have 1 and 1 different commits each, respectively.
(use "git pull" to merge the remote branch into yours)
```

Git push origin main

Git checkout

```
[hint: see the 'Note about fast-forwards' in 'git push --help' for details.]
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git push origin snow
Total 0 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'snow' on GitHub by visiting:
remote:   https://github.com/anubodhale/devops/pull/new/snow
remote:
To https://github.com/anubodhale/devops
 * (new branch)   snow -> snow
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git checkout
Your branch and 'origin/main' have diverged,
and have 1 and 1 different commits each, respectively.
(use "git pull" to merge the remote branch into yours)
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git checkout snow
Switched to branch 'snow'
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git status
On branch snow
nothing to commit, working tree clean
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$
```

Create new code or file

Git add

Git commit -m “added a file”

```
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git add .
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git commit -m "cc"
> ^C
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git commit -m "cc"
[main b0384ed] cc
1 file changed, 3 insertions(+), 4 deletions(-)
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ ls
hello1.py hello1.ty README.md
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git status
On branch main
Your branch and 'origin/main' have diverged,
```

Git pull origin main

```
dyp@dyp-OptiPlex-3000:~$ mkdir AsB_43
mkdir AsB_43: command not found
dyp@dyp-OptiPlex-3000:~$ cd AsB_43
dyp@dyp-OptiPlex-3000:~/AsB_43$ cd devoops
bash: cd: devoops: No such file or directory
dyp@dyp-OptiPlex-3000:~/AsB_43$ cd devops
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.02 KiB | 1.02 MiB/s, done.
From https://github.com/anubodhale/devops
 * branch          main      -> FETCH_HEAD
   7f10e54..bc115b0  main      -> origin/main
Updating 7f10e54..bc115b0
Fast-forward
 hello1.py | 7 ++++---
 1 file changed, 4 insertions(+), 3 deletions(-)
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 972 bytes | 108.00 KiB/s, done.
From https://github.com/anubodhale/devops
 * branch          main      -> FETCH_HEAD
   bc115b0..222a99a  main      -> origin/main
Updating bc115b0..222a99a
error: Your local changes to the following files would be overwritten by merge:
      hello1.py
Please commit your changes or stash them before you merge.
Aborting
```

Git push origin snow

Git checkout b branch name

```
hint: See the "Note about fast-forwards" in "git push --help" for details.
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git push origin snow
Total 0 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'snow' on GitHub by visiting:
remote:   https://github.com/anubodhale/devops/pull/new/snow
remote:
To https://github.com/anubodhale/devops
 * [new branch]      snow -> snow
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git checkout
Your branch and 'origin/main' have diverged,
and have 1 and 1 different commits each, respectively.
(use "git pull" to merge the remote branch into yours)
dyp@dyp-OptiPlex-3000:~/AsB_43/devops$ git checkout snow
Switched to branch 'snow'
```